

Section 31: Databases

239. Databases Explained: SQL vs NoSQL

- Types of databases:
 - SQL Database:
 - Structure: Relational databases that use structured query language (SQL) for defining and manipulating data.
 - Schema: Rigid schema with tables, rows, and columns. The schema must be defined before data can be inserted.
 - Data Integrity: Strong ACID (Atomicity, Consistency, Isolation, Durability) compliance ensures reliable transactions and data integrity.
 - Scaling: Vertical scaling (adding more power to the existing machine) is more common.
 - Examples: MySQL, PostgreSQL, Microsoft SQL Server, Oracle Database.
 - Use Cases: Suitable for applications requiring complex queries, transactions, and consistency, such as financial systems and enterprise applications.
 - NoSQL Database:
 - Structure: Non-relational databases that can store and retrieve data in various formats (document, key-value, column-family, graph).
 - Schema: Flexible schema allows for dynamic and unstructured data, which can change over time.
 - Data Integrity: Generally, follows BASE (Basically Available, Soft state, Eventually consistent) principles, favouring flexibility and availability over strict consistency.
 - Scaling: Horizontal scaling (adding more machines to handle the load) is more natural and cost-effective.
 - Examples: MongoDB (document), Cassandra (column-family), Redis (key-value), Neo4j (graph).
 - Use Cases: Ideal for applications requiring high performance, scalability, and flexible data models, such as big data applications, content management systems, and real-time analytics.